Status	Finished
Started	Saturday, 17 October 2026, 4:18 PM
Completed	Saturday, 17 October 2026, 4:22 PM
Duration	3 mins 34 secs
Grade	**15.00** out of 15.00 (**100%**)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are combined to form a single sorted list
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question 2

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Bubble Sort

Question 3

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sort()

Question 4

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order

b. Bubble Sort cannot handle duplicate elements

c. Bubble Sort is less efficient for large lists

d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question 5

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items

b. Sorting is rarely used in programming

c. Sorting helps in finding duplicates quickly

d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question 6

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower

b. It increases the number of comparisons needed

c. It has no effect on the search operation

d. It allows for faster searching

The correct answer is: It allows for faster searching

Question 7

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element

b. The parts are ignored

c. Each part is sorted independently

d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question 8

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It only works on small datasets
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question 9

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Searching
- c. Merging
- d. Sorting

The correct answer is:
Complexity

Question 10

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Rearranging
- c. Searching
- d. Merging

The correct answer is: Sorting

Question 11

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question 12

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method

b. Dynamic programming

c. Divide and conquer

d. Backtracking

The correct answer is:
Divide and conquer

Question 13

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower

b. Decreases the efficiency of algorithms

c. Makes data harder to manage

d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question 14

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question 15

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Searching for data in a list
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format